

Autodesk® Scaleform®

Scaleform

/

Scaleform 4.4

Nate Mitchell

1.01

2012 6 5

著作権表示

Autodesk® Scaleform® 4.4

© 2014 Autodesk, Inc. All rights reserved. Except as otherwise permitted by Autodesk, Inc., this publication, or parts thereof, may not be reproduced in any form, by any method, for any purpose.

Certain materials included in this publication are reprinted with the permission of the copyright holder.

The following are registered trademarks or trademarks of Autodesk, Inc., and/or its subsidiaries and/or affiliates in the USA and other countries: 123D, 3ds Max, Algor, Alias, AliasStudio, ATC, AutoCAD LT, AutoCAD, Autodesk, the Autodesk logo, Autodesk 123D, Autodesk CAM 360, Autodesk Homestyler, Autodesk Inventor, Autodesk MapGuide, Autodesk Streamline, AutoLISP, AutoSketch, AutoSnap, AutoTrack, Backburner, Backdraft, Beast, BIM 360, Burn, Buzzsaw, CADmep, CAiCE, CAMduct, CFdesign, Civil 3D, Cleaner, Combustion, Communication Specification, Configurator 360™, Constructware, Content Explorer, Creative Bridge, Dancing Baby (image), DesignCenter, DesignKids, DesignStudio, Discreet, DWF, DWG, DWG (design/logo), DWG Extreme, DWG TrueConvert, DWG TrueView, DWGX, DXF, Ecotect, ESTmep, Evolver, FABmep, Face Robot, FBX, Fempro, Fire, Flame, Flare, Flint, FMDesktop, ForceEffect, FormIt, Freewheel, Fusion 360, Glue, Green Building Studio, Heidi, Homestyler, HumanIK, i-drop, ImageModeler, Incinerator, Inferno, InfraWorks, InfraWorks 360, Instructables, Instructables (stylized robot design/logo), Inventor, Inventor HSM, Inventor LT, Kynapse, Kynogon, LandXplorer, Lustre, MatchMover, Maya, Maya LT, Mechanical Desktop, MIMI, Mockup 360, Moldflow Plastics Advisers, Moldflow Plastics Insight, Moldflow, Moondust, MotionBuilder, Movimento, MPA (design/logo), MPA, MPI (design/logo), MPX (design/logo), MPX, Mudbox, Navisworks, ObjectARX, ObjectDBX, Opticore, Pipeplus, Pixlr, Pixlr-o-matic, Productstream, Publisher 360, RasterDWG, RealDWG, ReCap, ReCap 360, Remote, Revit LT, Revit, RiverCAD, Robot, Scaleform, Showcase, Showcase 360 ShowMotion, Sim 360, SketchBook, Smoke, Socialcam, Softimage, Sparks, SteeringWheels, Stitcher, Stone, StormNET, TinkerBox, ToolClip, Topobase, Toxik, TrustedDWG, T-Splines, ViewCube, Visual LISP, Visual, VRED, Wire, Wiretap, WiretapCentral, XSI.

All other brand names, product names or trademarks belong to their respective holders.

Disclaimer

THIS PUBLICATION AND THE INFORMATION CONTAINED HEREIN IS MADE AVAILABLE BY AUTODESK, INC. "AS IS." AUTODESK, INC. DISCLAIMS ALL WARRANTIES, EITHER EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE REGARDING THESE MATERIALS.

How to Contact Autodesk Scaleform:

Document	Mobile Game Kit Overview
Address	Autodesk Scaleform Corporation 6305 Ivy Lane, Suite 310 Greenbelt, MD 20770, USA
Website	www.scaleform.com
Email	info@scaleform.com
Direct	(301) 446-3200
Fax	(301) 446-3199

目次

1	はじめに	1
1.1	Starforce Battlement	2
1.2		4
2	概要	5
2.1		5
2.2		6
2.2.1		6
2.2.2	HUD	8
3	アーキテクチャ	10
3.1	Flash	10
3.2	ActionScript	10
3.2.1		11
3.2.2	ActionScript	15
3.2.3		16

1 _____

Scaleform Autodesk® Scaleform®

Starforce Battlement Scaleform
ActionScript 3

PC Mac

Linux iOS Android

iOS ActionScript 3
Adobe®

Flash® Scaleform

Flash Player Scaleform Adobe
Scaleform

- Starforce Battlement
1. UI Flash
Professional
 2. ActionScript 3 (FlashDevelop 4)
 3. .SWF Scaleform
 4. Scaleform GPU

1.1 Starforce Battlement

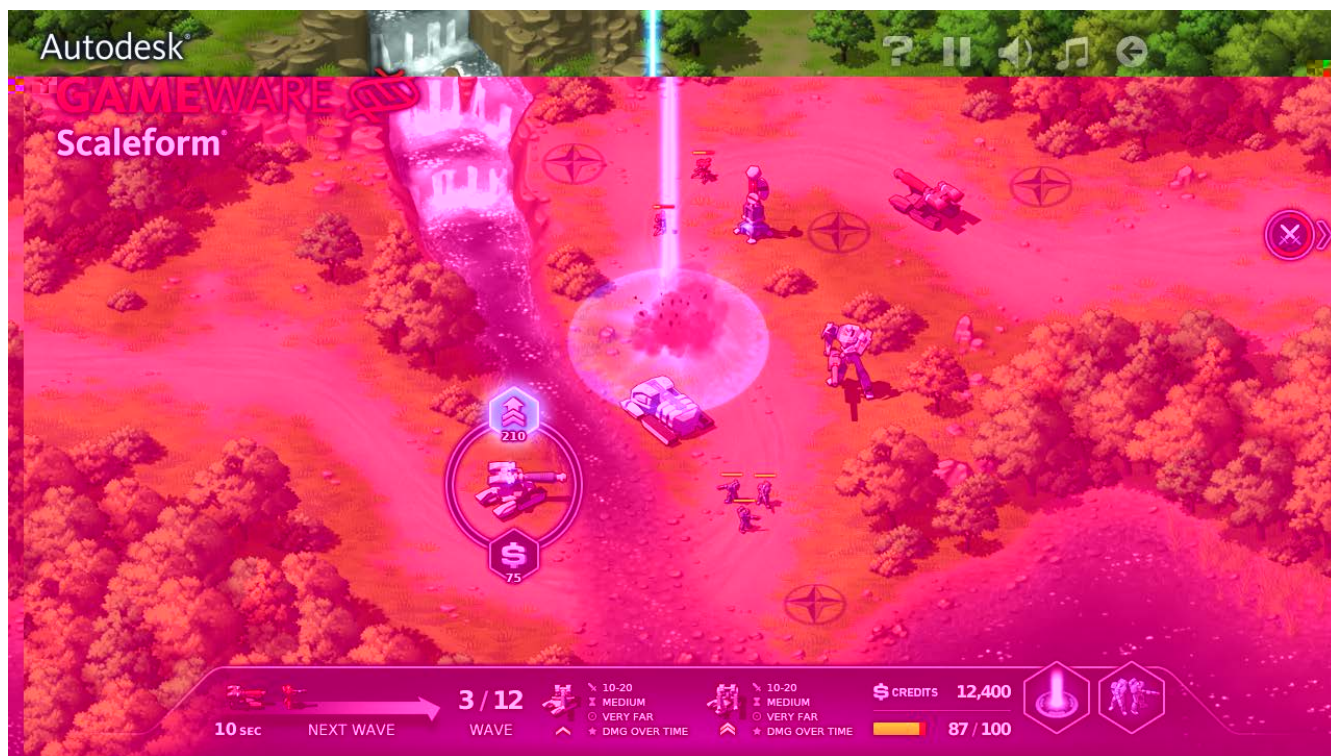


図 1 : Starforce – レベル 1 のスクリーンショット

Starforce Battlement

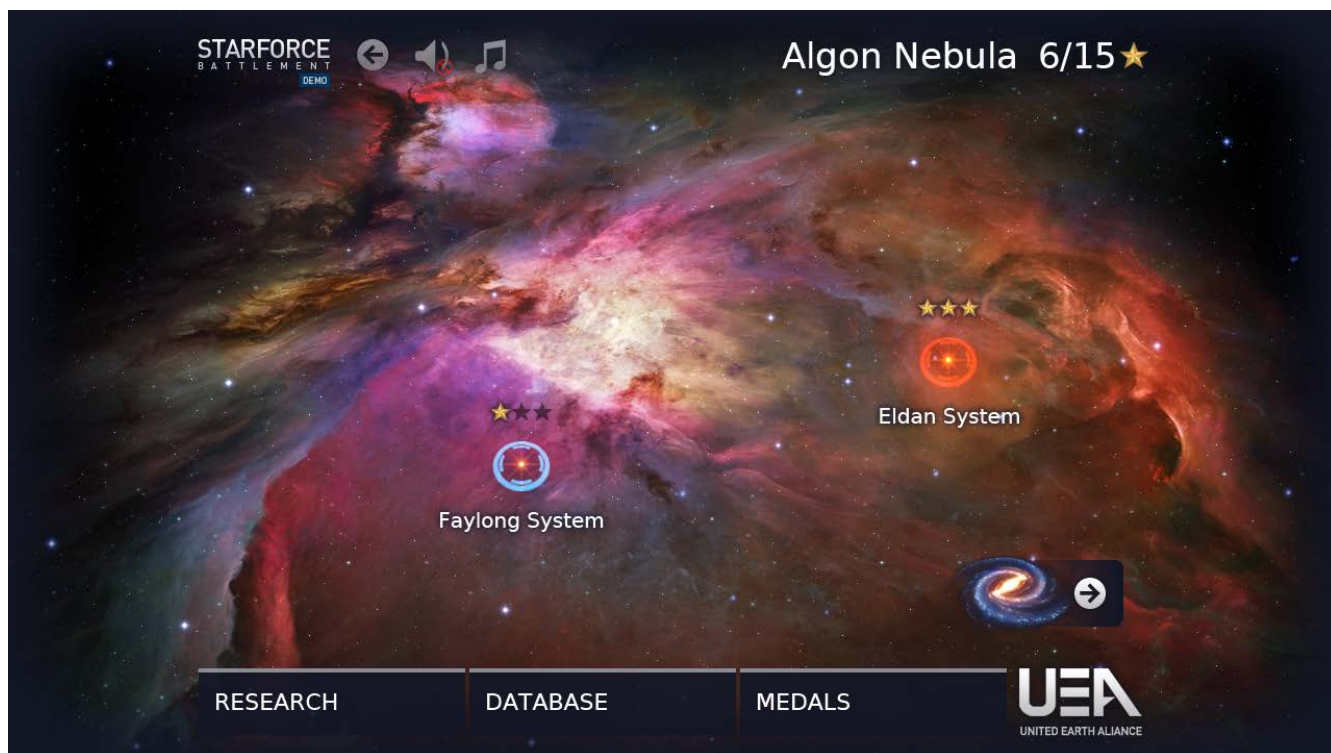


図 2 : Starforce - メインメニュー

Starforce Battlement

/

Starforce Battlement

iOS

2012

5

iOS App Store

Scaleform

iOS

1.2 Š

Starforce Battlement

Scaleform

- 1.
2. GPU
3. ActionScript 3
- 4.
- 5.
- 6.
7. iOS
- 8.

ActionScript 3

- 1.
- 2.
3. iOS
- 4.

2

2.1

x Bin/Data/AS3/Kits/StarforceTD/ - ActionScript 3 Flash

x Projects/Win32 /{Msvc80, Msvc90, = . Msvc10}/Gfx 4.4 SDK / – Windows
FxPlayer Visual Studio 2005/2008/2010 Visual Studio

x Starforce

x Projects/iPhone/Xcode4/Gfx 4. 4 iPhone SDK – iOS Xcode
4 Starforce

x LocalApps \ StarforceTD \ Android – Scaleform
Eclipse

Android StarforceTD

Windows StarforceTD.exe Bin/Data/AS3/Kits/ StarforceTD
Windows Scaleform SDK

FxPlayer FxMobilePlayer

Windows Bin/Data/AS3/Kits/StarforceTD
StarforceTD.swf

iOS

Xcode

SFW

iTunes

2.2 y” - ž ÷

2.2.1メインメニュー



図 3 : Starforce - メインメニュー

StarforceTD.as

StarforceTD.fla

MainMainState

StarforceMenu.swf

StarforceMenu.swf

ActionScript 3



図 4 : Starforce - ミッションデータ

MissionDataView.as

Enter Mission

X

2.2.2ゲーム開始と HUD



図 5 : Starforce – レベル 1 のゲーム画面と UI

Starforce Battlement

StarforceTD.swf

StarforceTD.swf

LevelGameState

5

/

() Rush Wave

HUD

1 3

1

HUD

2

HUD

Orbital Strike

Deploy Mercenaries

HUD

Orbital Strike

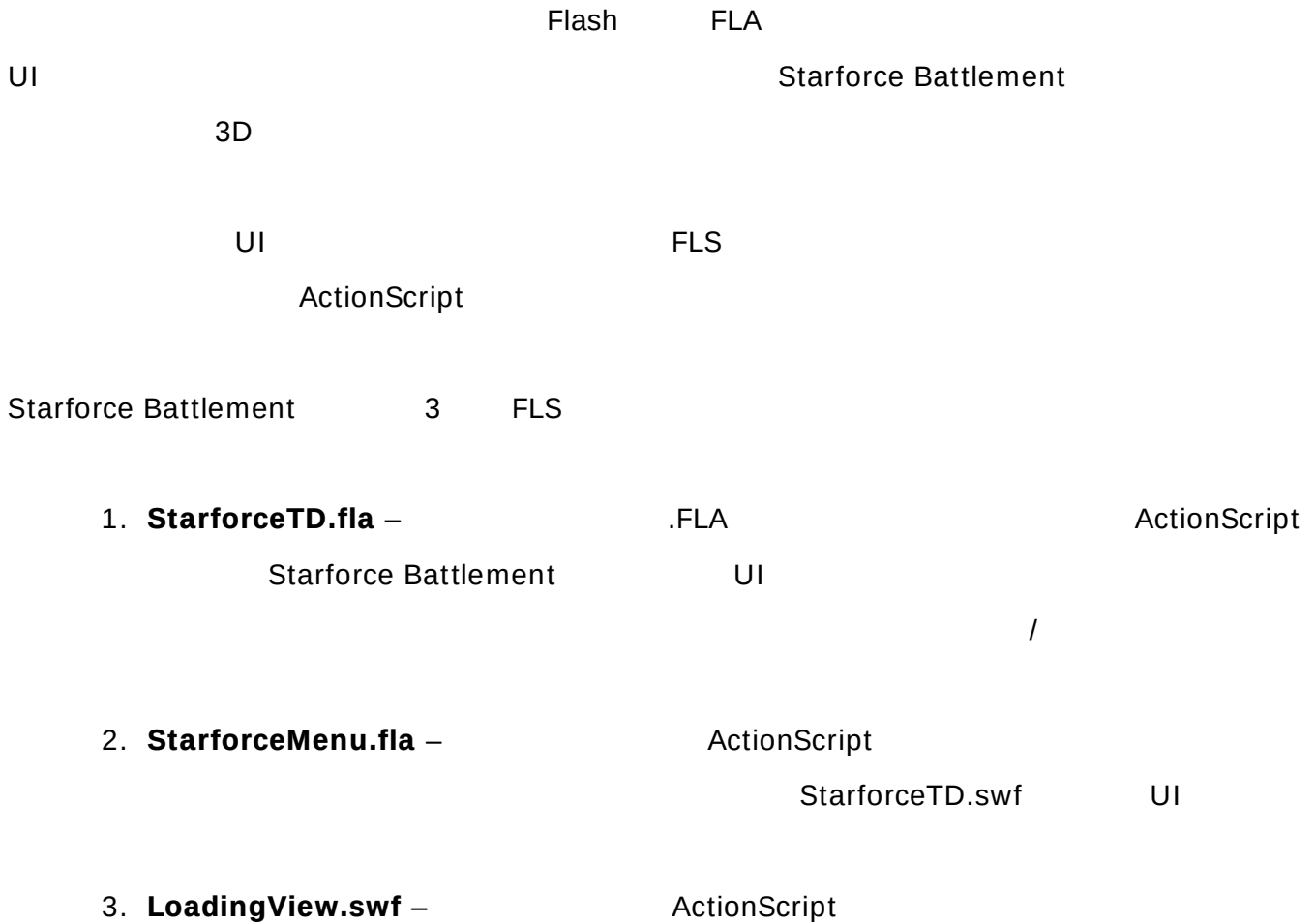
HUD

/

/

3 _____

3.1 Flash 3D UI •



3.2 ActionScript 4 {

3.2.1 実装サマリー

Starforce Battlement

UI

ActionScript 3

Starforce iOS

Starforce

3.2.1.1 エントリーポイント

StarforceTD.as

Flash

StarforceTD.swf

Adobe

Flash

3.2.1.2 ゲームステータス

3 GameStates

GameStates 1

GameStates

StarforceTD	3
-------------	---

1. MenuGameState – GameState
2. LoadingGameState – GameState
3. LevelGameState – GameState

StarforceTD

StarforceMenu.swf MainMenuView.as

3.2.1.3 メインメニューの実装

com.scaleform.std.menu

ActionScript 3

GameEvent

StarforceTD.as

LoadingGameState

MenuGameState

LevelGameState

LevelGameState

LoadingGameState

StarforceTD.swf

LoadingGameState

HUD

3.2.1.4 ゲームプレイの実装

Level.as

Level.as

HUD

Level

Level

Flash Professional

ActionScript

ActionScript

ActionScript

Starforce Battlement

ActionScript

Adobe Flash

TowerMech

StarforceTD.flas Flash

TowerMech

com.scaleform.std.entities.TowerMech ActionScript

ActionScript

Level.as

UI

HUD

AttackEvent

Level

AttackEvents

AttackEvent

3.2.1.5 ゲームデータ

com.scaleform.std.data

GameData

GameData

ActionScript

XML

JSON

XML

Scaleform 4.0

3.2.1.6 インプット処理

ActionScript

Scaleform Player

/

ActionScript

Scaleform

FxPlayer

MultitouchInputState

MovieView

Starforce

TouchEvents

MouseEvents

TouchEvents

MouseEvents

TouchEvent.TOUCH_TAP MouseEvent.MOUSE_CLICK
 TouchEvent.PRESS MouseEvent.MOUSE_DOWN

TouchEvent MouseEvent 1
 Event target

 is

```
protected function onSetRallyClick( e:Event ):void {
    var point:Point;
    if (e is MouseEvent) {
        var me:MouseEvent = e as MouseEvent;
        point = new Point( me.stageX, me.stageY );
    }
    else {
        var te:TouchEvent = e as TouchEvent;
        point = new Point( te.stageX, te.stageY );
    }
}
```

3.2.2 ActionScript パッケージ

ActionScript

ActionScript

com.scaleform.std

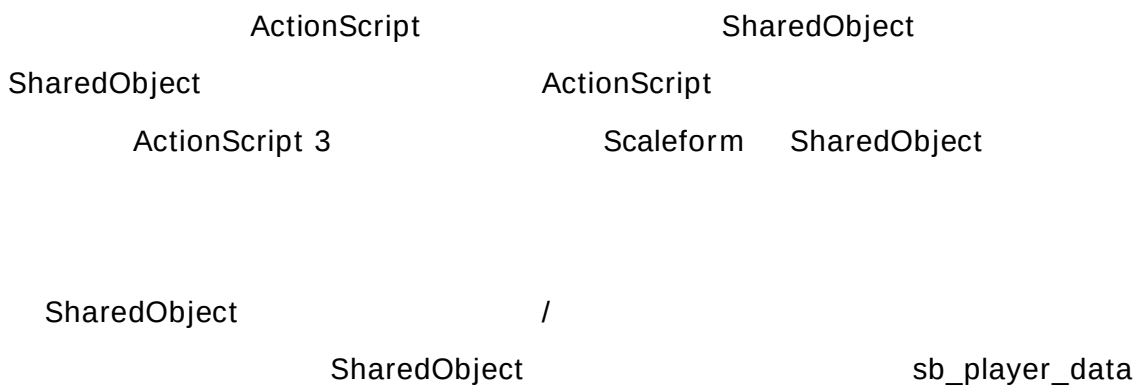
1. **abilities** – (Orbital Strike, Hire Mercenaries)
2. **controls** – UI
3. **core** – /
4. **data** –

- 5. entities –
- 6. events– UI
- 7. fx –
- 8. hud– HUD
- 9. levels – Level 1
- 10. loading –
- 11. menu – UI
- 12. system – /
ActionScript
- 13. utils – k-

3.2.3 フレームワーク

3.2.3.1 保存/ロードフレームワーク

com.scaleform.std.system.SaveSystem.as



```

// Local reference to the galaxy progress.                Populated from the PlayerData.
public static var galaxyProgress:Array = null;
// Constant String for the property of the SharedObject we'll modify.
protected static const GALAXY_PROGRESS_SO_PROP:String = "galaxyProgress";
// Reference to the SharedObject once we retrieve it.
protected static var _playerDataSO:SharedObject = null;

public static function load( pd:PlayerData ):void        {
    _playerDataSO = SharedObject.getLocal("sb_player_data");
    galaxyProgress = _playerDataSO.data[ GALAXY_PROGRESS_SO_PROP ];
    pd.setGalaxyProgress( galaxyProgress );
}

public static function save( pd:PlayerData ):void        {
    // Write PlayerData into SharedObject.
    _playerDataSO = SharedObject.getLocal("sb_player_data");
    galaxyProgress = pd.getGalaxyProgress();
    _playerDataSO.data[ GALAXY_PROGRESS_SO_PROP ] = galaxyProgress;
    _playerDataSO.flush();
}

```

1 SharedObject

SharedObjects

SharedObject

http://help.adobe.com/en_US/FlashPlatform/reference/actionscript/3/flash/net/SharedObject.html

3.2.3.2 サウンドフレームワーク

com.scaleform.std.system.SoundSystem

SoundEvents

MP3

SoundEvents

```

dispatchEvent( new SoundEvent( SoundEvent.PLAY_SOUND, true, true,
"infantry_attack.mp3",
1, false ) );

```

2

SoundEvents

BGM

SoundEvent

```
dispatchEvent( new SoundEvent( SoundEvent.MUTE_SOUND FX, true, true );
```